Camera images

Frontend

Image pyramid

Feature tracking

Feature recall

Feature detection (cam 0)

Feature matching

Overlapping mask generation

Feature detection (cam i > 0)
on non overlapping area

Filter points with epipolar constraint

Keypoints

Landmarks projection

Patch matching

Feature patches

IMU samples

IMU integration

Backend

VIO

Landmarks

Depth guess

Latest pose estimate

Sliding window poses estimate

Landmark database

Triangulation of new landmarks

Add covisible window

Place recognition trigger

Save map

Marginalization

Optimization

Take keyframe

Take long term keyframe

Add observations
for existing landmarks

Map Database

Read thread

Covisibility query

Island query

Write thread

Map stamp

Loop detection

Marginalization Stamp

Loop closure

Loop Closing

HashBoW indexing

HashBoW query

Consistency validation

Covisibility check

Island matching

Geometric validation

Matching

Build island

Landmarks projection

Geometric validation

Send loop detection

Pose graph optimization

Full BA

Load Problem

Allocate Landmark Blocks

Levenberg-Marquardt

1- Linearize

2- Marginalize landmarks

3- Solve RCS

4- Backsubstitute & update

5- Accept/reject step

Output kf poses